Web Development Activity Based Learning

Project Title: Adventure and Trekking Website

Description

This project is a single-page website for TrailTrek, a fictional trekking and camping company, created using HTML and CSS. The site presents popular trekking routes with their difficulty levels and durations, camping trips, a day-wise trip schedule, an equipment checklist, safety guidelines and package prices. A dark image overlay on the hero banner keeps the text readable over a photograph, and visually distinct information panels such as cards, a table, a checklist box and a safety panel organise the content so that it is clear and easy to scan.

1. Introduction

This activity introduces the two core technologies of the web: **HTML** (HyperText Markup Language), which defines the content and structure of a page, and **CSS** (Cascading Style Sheets), which controls how that content looks. The project is a single-page website for a fictional trekking company called **TrailTrek**. It presents everything a visitor would want to know before booking a trek: routes, camping options, the trip schedule, what to carry, safety rules and package prices.

Objectives

- To build a complete multi-section web page using semantic HTML tags.
- To style the page using CSS properties such as colours, spacing, borders, shadows and layout.
- To use an image overlay (a dark gradient over a background photo) so that text stays readable on a picture.
- To create visually distinct information panels such as cards, tables, a checklist box and a warning panel.
- To connect sections with internal navigation links and to provide an enquiry link for each package.

Website Sections at a Glance

Section	Purpose
Header and Navigation	Shows the brand name and tagline and links to every main section.
Hero Section	Large background image with a dark overlay, a heading, and a call-to-action button.
Popular Trekking Routes	Three cards giving the difficulty level and duration of each route.
Camping Trips	Two panels describing the weekend and adventure camping options.
Trip Schedule	A table of days, activities and timings.
Equipment Checklist	A highlighted list of items a trekker should carry.
Safety Guidelines	A warm-coloured panel with five safety rules.
Trekking Packages	Three price cards with enquiry buttons.
Footer	Closing message, contact e-mail and copyright line.

2. HTML Source Code (trekking.html)

The complete source file is listed below. The file keeps its CSS inside the **style** attribute of each tag (called *inline CSS*), so the HTML and the styling appear together.

Note: emoji characters in the original file are shown here as HTML character codes (for example 🌿 stands for the leaf emoji), because the PDF code font cannot draw emoji. The browser displays them as emoji.

Source Code in the Editor

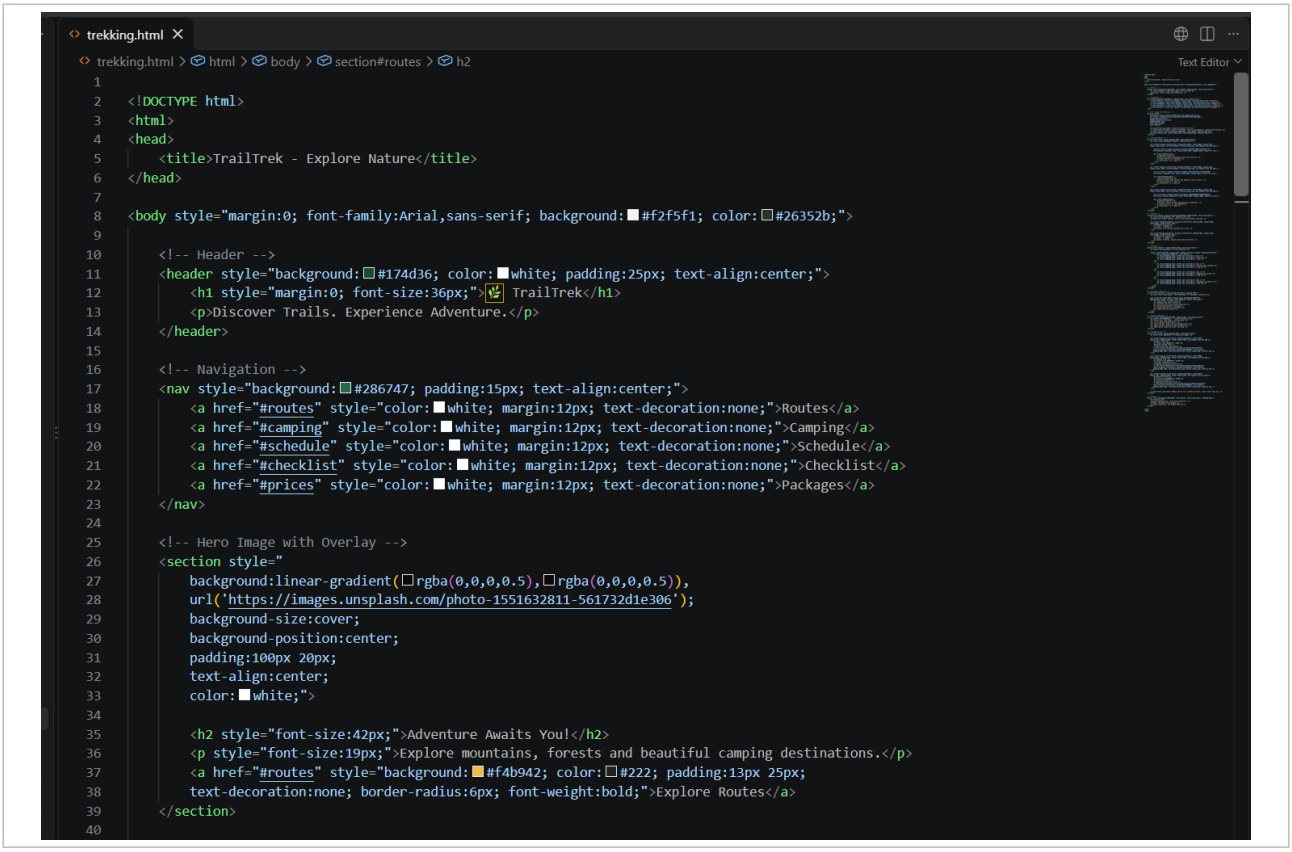


Figure 1: trekking.html in the code editor - head, header, navigation and hero section (lines 1-40)

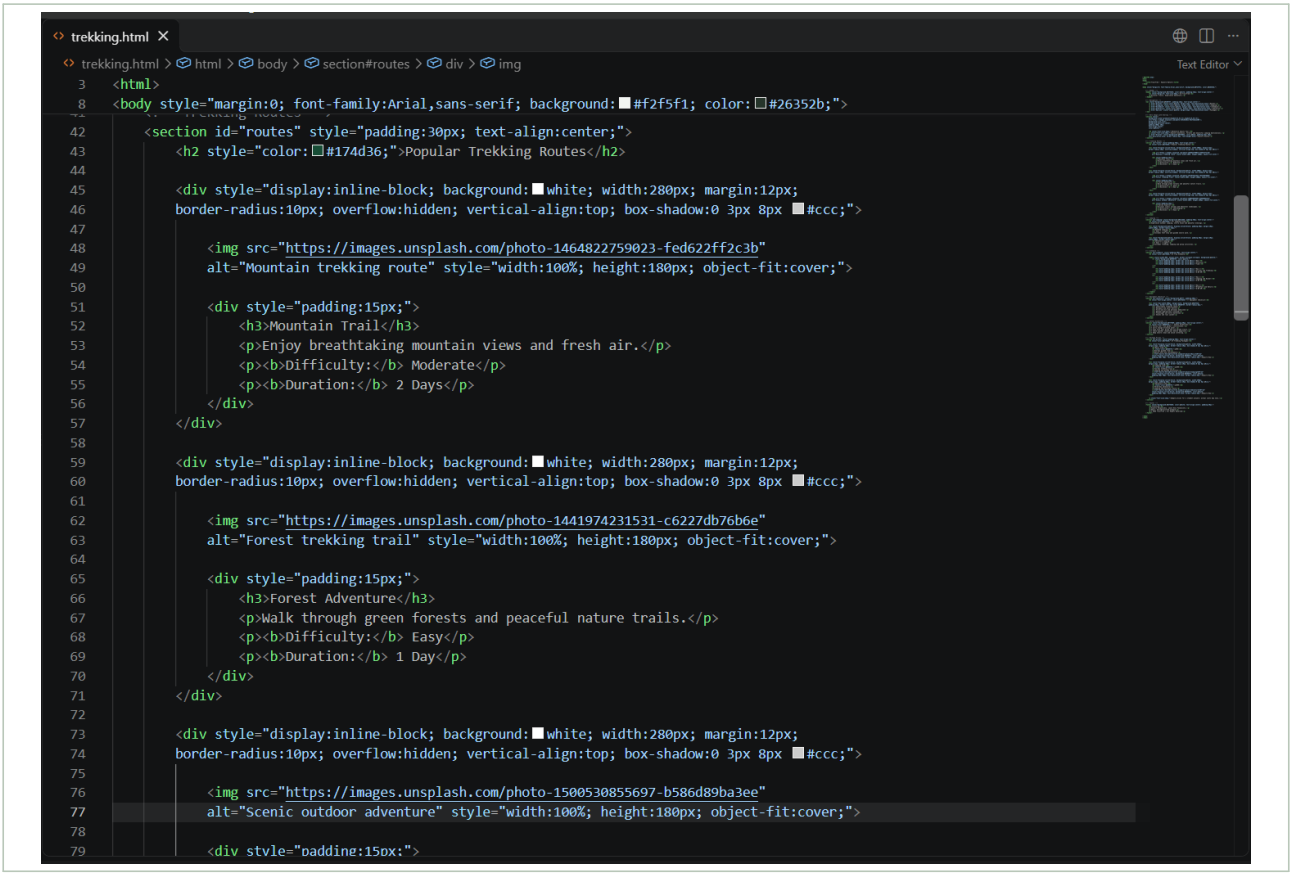


Figure 2: trekking.html in the code editor - body style and the Popular Trekking Routes section (lines 42-79)

Complete Source Listing

```
<!DOCTYPE html>
<html>
<head>
    <title>TrailTrek - Explore Nature</title>
</head>

<body style="margin:0; font-family:Arial,sans-serif; background:#f2f5f1; color:#26352b;">

    <!-- Header -->
    <header style="background:#174d36; color:white; padding:25px; text-align:center;">
        <h1 style="margin:0; font-size:36px;">&#127807; TrailTrek</h1>
        <p>Discover Trails. Experience Adventure.</p>
    </header>

    <!-- Navigation -->
    <nav style="background:#286747; padding:15px; text-align:center;">
        <a href="#routes" style="color:white; margin:12px; text-decoration:none;">Routes</a>
        <a href="#camping" style="color:white; margin:12px; text-decoration:none;">Camping</a>
        <a href="#schedule" style="color:white; margin:12px; text-decoration:none;">Schedule</a>
        <a href="#checklist" style="color:white; margin:12px; text-decoration:none;">Checklist</a>
        <a href="#prices" style="color:white; margin:12px; text-decoration:none;">Packages</a>
    </nav>

    <!-- Hero Image with Overlay -->
    <section style="
        background:linear-gradient(rgba(0,0,0,0.5),rgba(0,0,0,0.5)),
        url('https://images.unsplash.com/photo-1551632811-561732d1e306');
        background-size:cover;
        background-position:center;
        padding:100px 20px;
        text-align:center;
        color:white;">

        <h2 style="font-size:42px;">Adventure Awaits You!</h2>
        <p style="font-size:19px;">Explore mountains, forests and beautiful camping destinations.</p>
        <a href="#routes" style="background:#f4b942; color:#222; padding:13px 25px;
```

```
    text-decoration:none; border-radius:6px; font-weight:bold;">Explore Routes</a>
</section>

<!-- Trekking Routes -->
<section id="routes" style="padding:30px; text-align:center;">
    <h2 style="color:#174d36;">Popular Trekking Routes</h2>

    <div style="display:inline-block; background:white; width:280px; margin:12px;
    border-radius:10px; overflow:hidden; vertical-align:top; box-shadow:0 3px 8px #ccc;">

        <img src="https://images.unsplash.com/photo-1464822759023-fed622ff2c3b"
        alt="Mountain trekking route" style="width:100%; height:180px; object-fit:cover;">

        <div style="padding:15px;">
            <h3>Mountain Trail</h3>
            <p>Enjoy breathtaking mountain views and fresh air.</p>
            <p><b>Difficulty:</b> Moderate</p>
            <p><b>Duration:</b> 2 Days</p>
        </div>
    </div>

    <div style="display:inline-block; background:white; width:280px; margin:12px;
    border-radius:10px; overflow:hidden; vertical-align:top; box-shadow:0 3px 8px #ccc;">

        <img src="https://images.unsplash.com/photo-1441974231531-c6227db76b6e"
        alt="Forest trekking trail" style="width:100%; height:180px; object-fit:cover;">

        <div style="padding:15px;">
            <h3>Forest Adventure</h3>
            <p>Walk through green forests and peaceful nature trails.</p>
            <p><b>Difficulty:</b> Easy</p>
            <p><b>Duration:</b> 1 Day</p>
        </div>
    </div>

    <div style="display:inline-block; background:white; width:280px; margin:12px;
    border-radius:10px; overflow:hidden; vertical-align:top; box-shadow:0 3px 8px #ccc;">

        <img src="https://images.unsplash.com/photo-1500530855697-b586d89ba3ee"
        alt="Scenic outdoor adventure" style="width:100%; height:180px; object-fit:cover;">

        <div style="padding:15px;">
            <h3>Valley Explorer</h3>
            <p>Discover scenic valleys and beautiful landscapes.</p>
            <p><b>Difficulty:</b> Challenging</p>
            <p><b>Duration:</b> 3 Days</p>
        </div>
    </div>
</section>

<!-- Camping -->
<section id="camping" style="background:#dcebdd; padding:30px; text-align:center;">
    <h2 style="color:#174d36;">&#127957;&#65039; Camping Trips</h2>
    <p>Experience outdoor camping, starry skies and peaceful evenings.</p>

    <div style="background:white; display:inline-block; padding:20px; margin:10px;
    width:250px; border-radius:10px;">
        <h3>Weekend Camping</h3>
        <p>2 days / 1 night</p>
        <p>Includes tent stay and guided nature walk.</p>
    </div>

    <div style="background:white; display:inline-block; padding:20px; margin:10px;
    width:250px; border-radius:10px;">
        <h3>Adventure Camping</h3>
        <p>3 days / 2 nights</p>
        <p>Includes trekking, camping and group activities.</p>
    </div>
</section>

<!-- Schedule -->
<section id="schedule" style="padding:30px; text-align:center;">
    <h2 style="color:#174d36;">&#128197; Trip Schedule</h2>

    <table style="width:90%; margin:auto; border-collapse:collapse; background:white;">
        <tr style="background:#286747; color:white;">
```

```
            <th style="padding:12px; border:1px solid #ccc;">Day</th>
            <th style="padding:12px; border:1px solid #ccc;">Activity</th>
            <th style="padding:12px; border:1px solid #ccc;">Time</th>
        </tr>
        <tr>
            <td style="padding:12px; border:1px solid #ccc;">Day 1</td>
            <td style="padding:12px; border:1px solid #ccc;">Arrival and Trekking</td>
            <td style="padding:12px; border:1px solid #ccc;">8:00 AM</td>
        </tr>
        <tr>
            <td style="padding:12px; border:1px solid #ccc;">Day 1</td>
            <td style="padding:12px; border:1px solid #ccc;">Camping and Dinner</td>
            <td style="padding:12px; border:1px solid #ccc;">6:00 PM</td>
        </tr>
        <tr>
            <td style="padding:12px; border:1px solid #ccc;">Day 2</td>
            <td style="padding:12px; border:1px solid #ccc;">Breakfast and Return</td>
            <td style="padding:12px; border:1px solid #ccc;">8:00 AM</td>
        </tr>
    </table>
</section>

<!-- Equipment Checklist -->
<section id="checklist" style="background:#fff; padding:30px;">
    <h2 style="text-align:center; color:#174d36;">&#127890; Equipment Checklist</h2>

    <div style="max-width:500px; margin:auto; background:#edf4eb;
    padding:20px; border-left:6px solid #286747; border-radius:8px;">
        <p>☑ Comfortable trekking shoes</p>
        <p>☑ Backpack and water bottle</p>
        <p>☑ First-aid kit and personal medicines</p>
        <p>☑ Weather-appropriate clothing</p>
        <p>☑ Flashlight and extra batteries</p>
        <p>☑ Snacks and rain jacket</p>
    </div>
</section>

<!-- Safety Guidelines -->
<section style="background:#fff0d9; padding:30px; text-align:center;">
    <h2 style="color:#98551b;">⚠&#65039; Safety Guidelines</h2>
    <p>✔ Follow your trek guide's instructions.</p>
    <p>✔ Carry enough drinking water.</p>
    <p>✔ Stay with your group and use marked trails.</p>
    <p>✔ Check weather conditions before departure.</p>
    <p>✔ Keep nature clean and avoid littering.</p>
</section>

<!-- Package Prices -->
<section id="prices" style="padding:30px; text-align:center;">
    <h2 style="color:#174d36;">&#128176; Trekking Packages</h2>

    <div style="display:inline-block; background:white; width:250px;
    margin:12px; padding:20px; border-radius:10px; box-shadow:0 3px 8px #ccc;">
        <h3>Basic Trek</h3>
        <h2 style="color:#286747;">₹999</h2>
        <p>One-day guided trek</p>
        <p>Nature walk and refreshments</p>
        <a href="mailto:hello@trailtrek.example?subject=Basic%20Trek"
        style="display:inline-block; background:#286747; color:white;
        padding:10px 20px; text-decoration:none; border-radius:5px;">Enquire Now</a>
    </div>

    <div style="display:inline-block; background:white; width:250px;
    margin:12px; padding:20px; border-radius:10px; box-shadow:0 3px 8px #ccc;">
        <h3>Camping Trek</h3>
        <h2 style="color:#286747;">₹2499</h2>
        <p>Two-day trekking trip</p>
        <p>Camping and guided activities</p>
        <a href="mailto:hello@trailtrek.example?subject=Camping%20Trek"
        style="display:inline-block; background:#286747; color:white;
        padding:10px 20px; text-decoration:none; border-radius:5px;">Enquire Now</a>
    </div>

    <div style="display:inline-block; background:white; width:250px;
    margin:12px; padding:20px; border-radius:10px; box-shadow:0 3px 8px #ccc;">
        <h3>Adventure Trek</h3>
```

```
            <h2 style="color:#286747;">₹3999</h2>
            <p>Three-day adventure</p>
            <p>Camping and multiple trails</p>
            <a href="mailto:hello@trailtrek.example?subject=Adventure%20Trek"
            style="display:inline-block; background:#286747; color:white;
            padding:10px 20px; text-decoration:none; border-radius:5px;">Enquire Now</a>
        </div>

        <p style="font-size:13px;">Sample prices for a student project; actual costs may vary.</p>
    </section>

    <!-- Footer -->
    <footer style="background:#174d36; color:white; text-align:center; padding:20px;">
        <h3>TrailTrek</h3>
        <p>Explore Responsibly. Leave Only Footprints.</p>
        <p>Email: hello@trailtrek.example</p>
        <p>© 2026 TrailTrek | All Rights Reserved</p>
    </footer>

</body>
</html>
```

3. Description of the HTML Code

Document Structure

The file begins with **<!DOCTYPE html>**, which tells the browser to treat the file as modern HTML. The **<html>** element wraps the whole page. Inside it, **<head>** holds information about the page, here the **<title>** "TrailTrek - Explore Nature" that appears on the browser tab. The visible content is placed inside **<body>**. Comments such as *<!-- Header -->* label each section and make the code easier to read.

Tags Used and Their Roles

Tag	Where it is used	Purpose
<header>, <nav>, <section>, <footer>	Top banner, menu, each content block, bottom of page	Semantic tags that give meaning to the page layout.
<h1> to <h3>	Site name, section titles, card titles	Headings in decreasing order of importance.
<p> and <b>	Descriptions, difficulty and duration	Paragraph text and bold labels such as "Difficulty:".
<a href>	Menu, hero button, Enquire Now buttons	Links. "#routes" jumps to the section with id="routes"; "mailto:" opens the e-mail app.
id attribute	routes, camping, schedule, checklist, prices	Gives a section a unique name so links can target it.
<img>	Three route cards	Displays a photo; src gives the image address and alt gives a text description.
	Cards, checklist box	A general container used to group content into panels.
<table>, <tr>, <th>, <td>	Trip schedule	Rows, header cells and data cells for the Day / Activity / Time table.

Section-by-Section Explanation

- **Header:** contains the <h1> heading "TrailTrek" with a leaf emoji and a tagline in a <p> tag.
- **Navigation:** five links (Routes, Camping, Schedule, Checklist, Packages) that scroll to the matching section ids on the same page.
- **Hero section:** a <section> with a heading, a short line of text and an "Explore Routes" link that is styled like a button.
- **Routes:** three cards, each with an image, a <h3> title, a description, and the difficulty (Moderate, Easy, Challenging) and duration (2, 1 and 3 days).
- **Camping:** two panels, "Weekend Camping" (2 days / 1 night) and "Adventure Camping" (3 days / 2 nights).
- **Schedule:** a table with a header row and three data rows describing arrival, camping and the return journey.
- **Checklist:** six lines, each starting with a tick symbol, listing the equipment to carry.
- **Safety guidelines:** five short rules, each starting with a check mark.

- **Packages:** three cards (Basic Trek Rs. 999, Camping Trek Rs. 2499, Adventure Trek Rs. 3999), each with an "Enquire Now" mailto link whose subject line is pre-filled, followed by a note that prices are samples.
- **Footer:** brand name, a closing message, the contact e-mail and a copyright line.

4. CSS Code

In trekking.html all styling is written as inline CSS inside style="..." attributes. To make the styles easier to study, they are listed below as an equivalent external stylesheet, grouped section by section. The property names and values are the same as those used in the HTML file. Figures 1 and 2 in Section 2 show these inline styles inside the editor, for example the *background* and *color* values of the header, the *linear-gradient* overlay of the hero section and the card styles of the routes section.

```
/* Equivalent CSS rules for the inline styles used in trekking.html */

body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: #f2f5f1;
    color: #26352b;
}

/* ---------- Header and navigation ---------- */
header { background: #174d36; color: white; padding: 25px; text-align: center; }
header h1 { margin: 0; font-size: 36px; }

nav { background: #286747; padding: 15px; text-align: center; }
nav a { color: white; margin: 12px; text-decoration: none; }

/* ---------- Hero section with image overlay ---------- */
.hero {
    background: linear-gradient(rgba(0,0,0,0.5), rgba(0,0,0,0.5)),
                url('https://images.unsplash.com/photo-1551632811-561732d1e306');
    background-size: cover;
    background-position: center;
    padding: 100px 20px;
    text-align: center;
    color: white;
}
.hero h2 { font-size: 42px; }
.hero p  { font-size: 19px; }
.hero a  { background: #f4b942; color: #222; padding: 13px 25px;
           text-decoration: none; border-radius: 6px; font-weight: bold; }

/* ---------- Trekking routes ---------- */
#routes { padding: 30px; text-align: center; }
#routes h2 { color: #174d36; }
.route-card {
    display: inline-block; background: white; width: 280px; margin: 12px;
    border-radius: 10px; overflow: hidden; vertical-align: top;
    box-shadow: 0 3px 8px #ccc;
}
.route-card img { width: 100%; height: 180px; object-fit: cover; }
.route-card .info { padding: 15px; }

/* ---------- Camping ---------- */
#camping { background: #dcebdd; padding: 30px; text-align: center; }
#camping h2 { color: #174d36; }
.camp-card { background: white; display: inline-block; padding: 20px;
             margin: 10px; width: 250px; border-radius: 10px; }

/* ---------- Schedule table ---------- */
#schedule { padding: 30px; text-align: center; }
#schedule h2 { color: #174d36; }
table { width: 90%; margin: auto; border-collapse: collapse; background: white; }
tr.head { background: #286747; color: white; }
```

```
th, td { padding: 12px; border: 1px solid #ccc; }

/* ---------- Equipment checklist ---------- */
#checklist { background: #fff; padding: 30px; }
#checklist h2 { text-align: center; color: #174d36; }
.checklist-box {
    max-width: 500px; margin: auto; background: #edf4eb; padding: 20px;
    border-left: 6px solid #286747; border-radius: 8px;
}

/* ---------- Safety guidelines ---------- */
.safety { background: #fff0d9; padding: 30px; text-align: center; }
.safety h2 { color: #98551b; }

/* ---------- Package prices ---------- */
#prices { padding: 30px; text-align: center; }
#prices h2 { color: #174d36; }
.package-card {
    display: inline-block; background: white; width: 250px; margin: 12px;
    padding: 20px; border-radius: 10px; box-shadow: 0 3px 8px #ccc;
}
.package-card h2 { color: #286747; }
.package-card a {
    display: inline-block; background: #286747; color: white;
    padding: 10px 20px; text-decoration: none; border-radius: 5px;
}
.price-note { font-size: 13px; }

/* ---------- Footer ---------- */
footer { background: #174d36; color: white; text-align: center; padding: 20px; }
```

5. Description of the CSS Code

How CSS Is Written

A CSS declaration has the form **property: value;**, for example *color: white;*. In this project the declarations are placed in the **style** attribute of each tag, which means they affect only that element. When the same declarations are written in a stylesheet, a **selector** (such as *nav a* or *#routes*) chooses which elements they apply to.

Key Properties Used

Property	Used for	Example from the project
margin, padding	Space outside and inside an element	padding:100px 20px on the hero section; margin:12px on cards
background, color	Panel colour and text colour	Dark green #174d36 header with white text
font-family, font-size	Typeface and text size	Arial, sans-serif; 36px main heading
text-align	Aligns text	center for most sections
text-decoration:none	Removes link underline	Navigation links and buttons
border, border-left	Lines around or beside an element	Green 6px left border on the checklist box
border-radius	Rounded corners	10px on cards, 6px on the hero button
box-shadow	Soft shadow under a panel	0 3px 8px #ccc on route and package cards
display:inline-block	Places cards side by side while keeping width	Route, camping and package cards
width, max-width, height	Size of an element	Cards 280px / 250px wide; images 180px tall
object-fit:cover	Crops an image to fill its box without stretching	Route images
border-collapse	Merges table borders into single lines	Trip schedule table
overflow:hidden	Clips content to the rounded corners	Route cards

Image Overlay (Hero Section)

The hero section uses a **layered background**. The first layer is *linear-gradient(rgba(0,0,0,0.5), rgba(0,0,0,0.5))*, a black layer that is 50% transparent. The second layer is the photograph set with *url(...)*. Because the dark layer sits on top of the photo, the white heading and text remain easy to read. *background-size: cover* makes the photo fill the whole section and *background-position: center* keeps the middle of the photo in view.

Visually Distinct Information Panels

- **Colour coding:** each area has its own background (white cards, light-green camping section #dcebdd, pale-green checklist box #edf4eb, cream safety panel #fff0d9) so the reader can tell the sections apart at a glance.
- **Cards:** white boxes with rounded corners and a shadow look raised from the page.
- **Accent border:** the checklist box has a thick green left border to draw attention.

- **Warning colour:** the safety section uses a warm cream background with a brown heading (#98551b) to signal caution.
- **Buttons:** links with padding, background colour and rounded corners look like buttons: yellow for the main call to action, green for "Enquire Now".
- **Table styling:** a green header row with white text, and 12px padding with light-grey borders on every cell.

6. Output

When trekking.html is opened in a web browser, it appears as a single scrolling web page. The screenshots below show the page from the top down to the Safety Guidelines section.

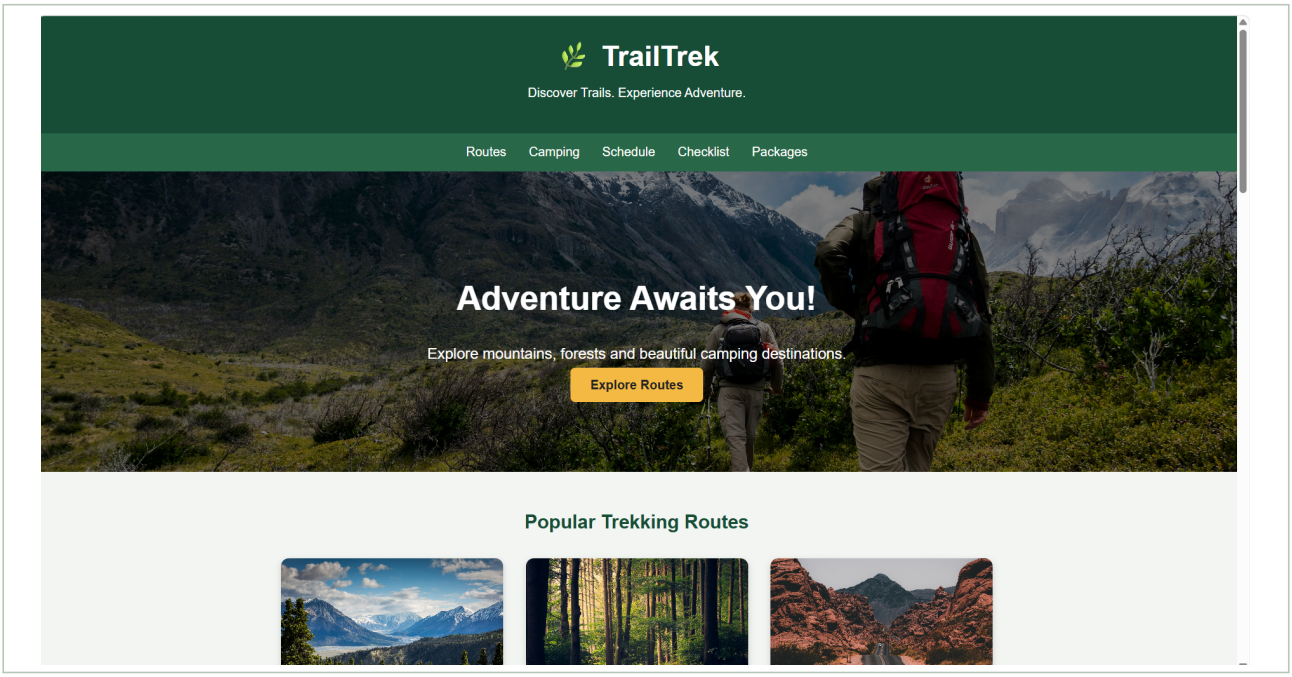


Figure 3: Output - header, navigation bar, hero section with image overlay, and the start of Popular Trekking Routes

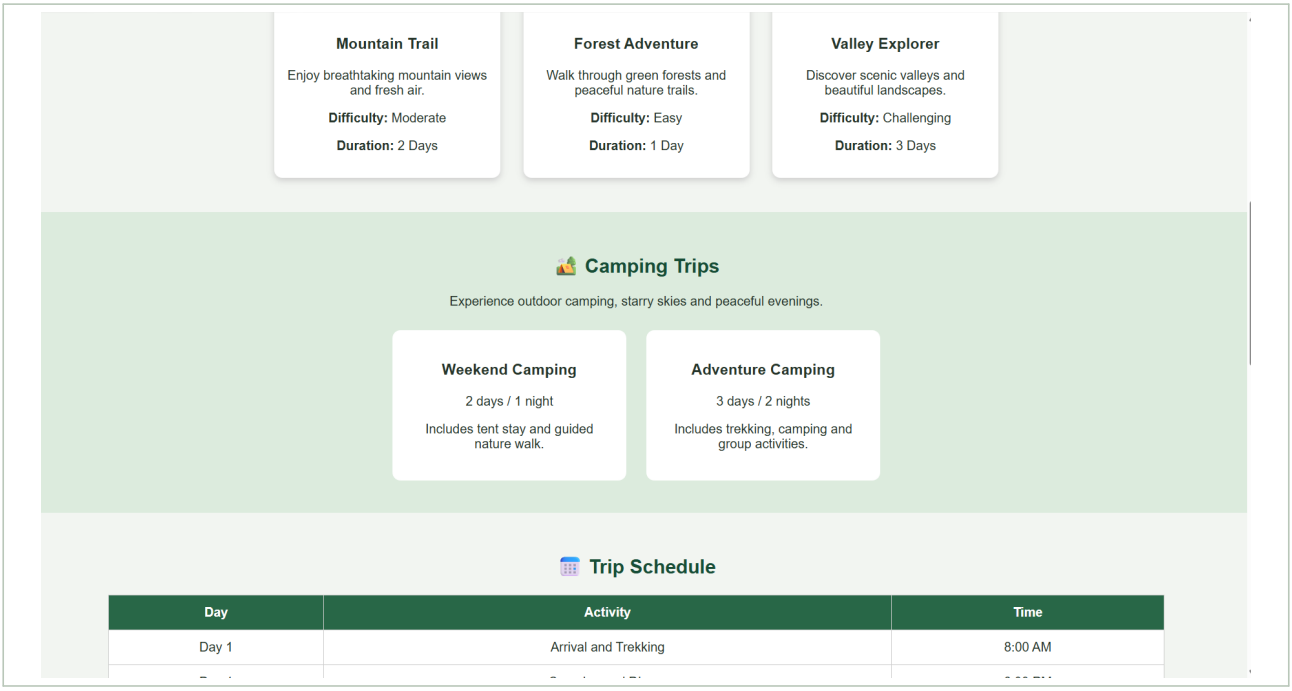


Figure 4: Output - route cards, Camping Trips section and the start of the Trip Schedule table

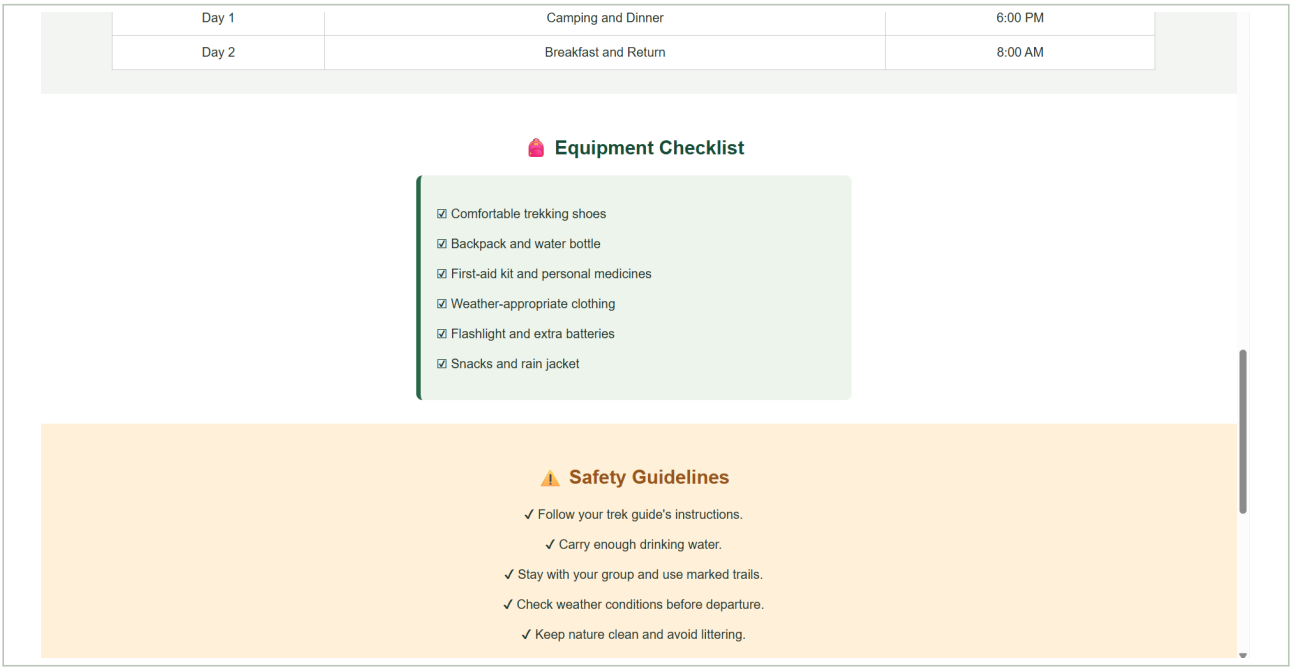


Figure 5: Output - end of the schedule table, Equipment Checklist and Safety Guidelines

Description of the Output

- **Header and navigation (Figure 3):** a dark green header shows the TrailTrek name with a leaf emoji and the tagline "Discover Trails. Experience Adventure." Below it, a lighter green bar holds the five centred links: Routes, Camping, Schedule, Checklist and Packages.
- **Hero section (Figure 3):** a full-width photograph of two hikers on a mountain trail is darkened by the overlay, so the white heading "Adventure Awaits You!" and the line below it are easy to read. A yellow "Explore Routes" button sits under the text.

- **Popular Trekking Routes (Figures 3 and 4):** three white cards with rounded corners and soft shadows, each with a photograph (mountains, forest, red-rock valley), a title, a short description, the difficulty (Moderate, Easy, Challenging) and the duration (2 Days, 1 Day, 3 Days).
- **Camping Trips (Figure 4):** a light-green panel with a camping emoji in the heading and two white boxes, Weekend Camping (2 days / 1 night) and Adventure Camping (3 days / 2 nights).
- **Trip Schedule (Figures 4 and 5):** a table with a green header row (Day, Activity, Time) and rows for Arrival and Trekking at 8:00 AM, Camping and Dinner at 6:00 PM, and Breakfast and Return at 8:00 AM, separated by thin grey borders.
- **Equipment Checklist (Figure 5):** six ticked items inside a pale-green box with a thick green left border, under a backpack emoji heading.
- **Safety Guidelines (Figure 5):** a cream-coloured section with a brown heading and warning emoji, listing five rules, each with a tick mark.

Below the safety guidelines, the page continues with the Trekking Packages section (three price cards with "Enquire Now" buttons) and the dark green footer with the contact e-mail and copyright line.

7. Conclusion

This activity resulted in a complete, well-organised website for a trekking company, built using only HTML and CSS. It showed how a plain document can be turned into a professional-looking page by dividing content into sections, linking them together, and styling each section so that it is easy to read and visually distinct. Working on the project connects theory with practice: every tag and property studied in class is used in a real, visible result.

Learning Outcomes

- Understood the basic structure of an HTML document (DOCTYPE, html, head, title, body).
- Used semantic tags (header, nav, section, footer) and headings, paragraphs, images, links, tables and containers (div) correctly.
- Created internal navigation using id attributes and anchor links, and an e-mail enquiry link using mailto.
- Applied CSS for colours, fonts, spacing, borders, rounded corners and shadows.
- Built an image overlay using a layered background with linear-gradient and rgba transparency.
- Designed visually distinct information panels (cards, table, checklist box, warning panel) and used colour to separate content.
- Learned to present information clearly: routes, difficulty levels, schedules, equipment, safety rules and prices.
- Practised writing alt text for images and structuring content for accessibility and readability.
- Understood the difference between inline CSS and an external stylesheet, and why separating them is useful for larger projects.

Overall, the activity strengthened problem-solving, attention to design detail and the ability to turn a written requirement into a working web page, which are skills that form the foundation for further study in web development and software engineering.